



Wastewater Business Plan

Lift · ATU · WWTP · Phase 1 Cloud · 2026

A company powered by **The Purple Standard**
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Vertical business plan · Version 1.1

Document generated: 2026-08-09

Phase 1 SaaS · 500-device fleet · cellular cost model

Executive Summary

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mooreVIEW — Wastewater Business Plan

Lift stations · ATU · WWTP · Phase 1 Cloud SaaS · 2026

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Executive summary

mooreVIEW enters the **US wastewater and onsite treatment market** as a **field-service monitoring platform** — proactive lift-station, ATU/ATS, and WWTP head-end monitoring with portable `.est` projects, PdM → CMMS, and MQTT Parc fleet uplink to **Phase 1 Cloud SaaS**.

Metric	Year 1	Year 2	Year 3
Total revenue (directional)	~\$550K	~\$1.74M	~\$3.9M
Cumulative monitored assets	~230	~380	~500+
End-of-year renewal ARR	~\$55K	~\$175K	~\$400K

Phase 1 cloud stack (SaaS + MQTT + Archive + Mongo) supports the full **500-device wastewater fleet** on a single NYC1 topology. At steady state:

Category	Monthly	Annual
Renewal revenue (500 assets, list)	~\$10,750	~\$129,000
DO Phase 1 cloud stack	~\$97	~\$1,167
Cellular — 500 field assets (\$2 SIM + \$0.50/50 MB)	\$1,250	\$15,000
Cellular — 32 WWTP/head-end appliances (\$2 SIM + \$5/1 GB)	\$224	\$2,688
Total infrastructure	~\$1,571	~\$18,855
Contribution after infra	~\$9,179	~\$110,145
Infra as % of renewal revenue	~14.6%	~14.6%

Cloud alone is ~0.9% of renewal revenue at 500 assets. **Cellular backhaul** is the dominant opex line (~11.6% of renewal at cost; billed to customer at **\$3/mo** list = **\$1,500/mo** revenue vs **\$1,250/mo** cost → **\$250/mo** cell margin).

Serviceable market: ~\$400M/yr US wedge (sites without full SCADA). **5-year ARR ceiling (wastewater vertical):** **\$15 – 25M** recurring.

Beachhead channel: Purple Standard sister operators (**ACE Septic & Waste**) — reference fleet, install crews, first **100–500 sites** without cold national sales.

Part 1 — Market & product

Vertical scope

Segment	Asset	Proactive wedge	Project pack	Tier
Lift stations	Duplex/triplex pump stations	Level/pump trends before SSO	duplex-lift-station	Commercial
ATU / ATS	Residential & commercial treatment	Blower/UV/pump health; compliance route	ATS Compliance Pack	Res / Com
WWTP head-end	Small municipal / package plants	Pump/blower PdM; remote operator view	mle-wastewater	Commercial
Septic OEM	Panel-builder embed	Provider fleet Parc	ATS.est	OEM

Who buys

Buyer	Wastewater use case
Service contractors (ACE-class)	Monitoring contracts on lift + ATU routes
Utilities / collections districts	Fleet Parc; fewer SSO events; 13 mo compliance archive
OEM panel shops	Connected ATS panel vs dialer-only skid
Property managers / commercial	ATU compliance before inspection failure

vs legacy

Solution	Year-1 TCO (duplex lift)	PdM → CMMS
Autodialer	~\$600 – \$1,200	○
mooreVIEW (+ cell)	~\$2,836	●
VTScada (1K I/O)	~\$7K – \$12K	○

List pricing (wastewater assets)

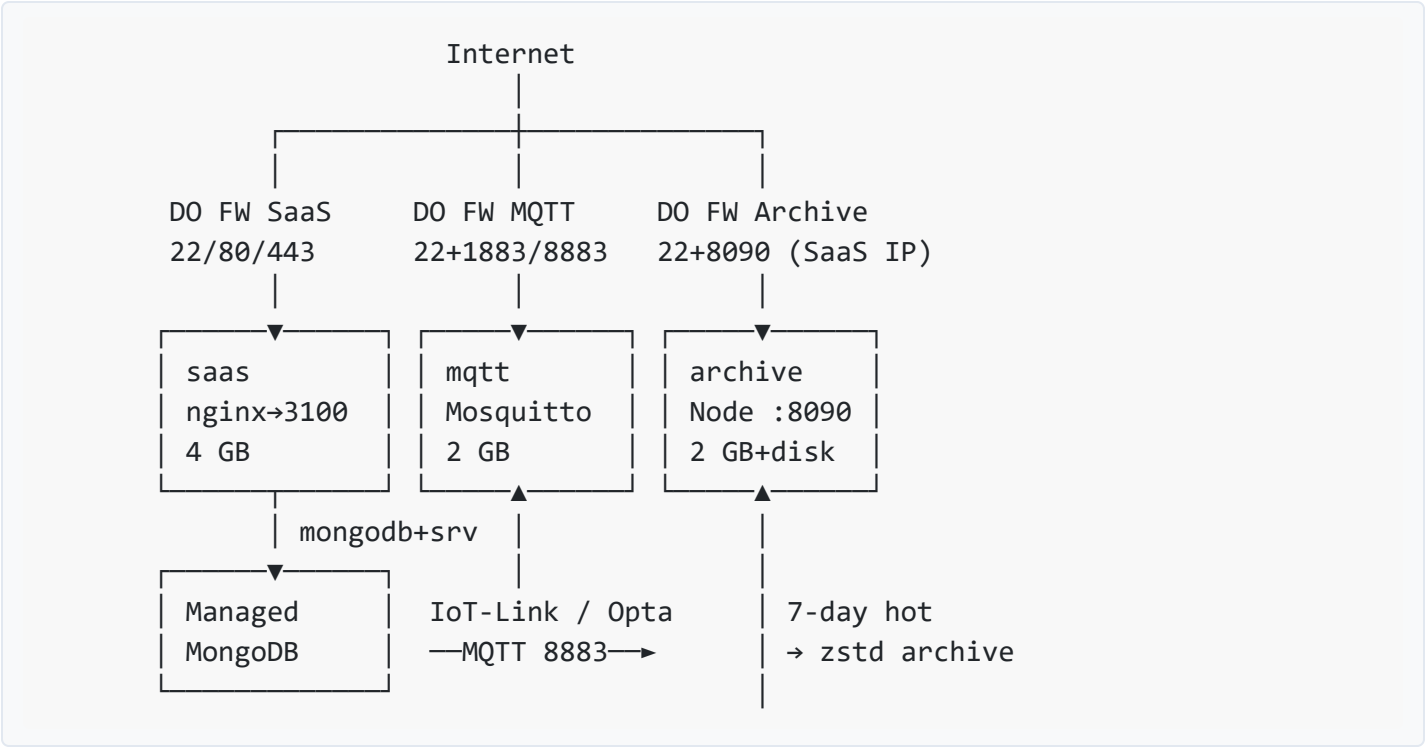
Asset	One-time	Renewal/mo	Cell (list)	Year-1 total
ATU — residential	\$1,500	\$18	\$3	\$1,752
ATU — commercial	\$2,500	\$25	\$3	\$2,836

Lift station — duplex	\$2,500	\$25	\$3	\$2,836
Well & pump (adjacent)	\$2,500	\$25	\$3	\$2,836

Renewal includes Parc uplink, historian, CMMS, PdM, alarm profiles, platform updates.

Part 2 — Phase 1 Cloud SaaS platform

Production topology (NYC1)



Host	Role	Size
mooreview-saas	nginx → Cloud Studio :3100	4 GB / 2 vCPU
mooreview-mqtt	Mosquitto TLS :8883	2 GB / 1 vCPU
mooreview-archive	Cold historian API :8090	2 GB + block volume
Managed MongoDB	Hot historian (7-day rolling)	10–20 GB tier

Edge path: IoT-Link or Arduino Opta (cellular) → `mqtt://mqtt.mooreview.io:8883` → SaaS Parc ingest → Mongo hot → day-7 zstd export to archive.

WWTP head-ends (32 units): MVP Suite appliance on **3090** with site agent + optional cameras; cellular uplink for inventory sync and remote operator view — credentials and LAN I/O stay on-site.

Part 3 — Capacity limits @ wastewater fleet

Platform maximums (Phase 1 single stack)

Resource	Phase 1 max	Wastewater @ 500 assets	Headroom
Devices per tenant	1,000	500	50%
Sites / locations	1,000	~120 contractors + 32 WWTP	87%
Tags (MOOREVIEW_MAX_TAGS)	70,912 (cloud default)	~ 7,500 est.	89%
MQTT connections	~2,000	532 (500 + 32)	73%
Hot Mongo (7-day)	15 GB base plan	~ 6.5 GB	fits base tier

Tag budget by asset type

Asset type	Typical tags/site	PdM pens	Sample rate
Duplex lift station	18 – 28	Pump runtime, starts, amp imbalance	5 s (PdM)
ATU — residential	8 – 12	Blower, UV hours, high water	5 – 30 s
ATU — commercial	12 – 18	Blower, UV, pumps, compliance	5 s
WWTP head-end (appliance)	40 – 120	Blower, clarifier, pumps (local)	Rollups to cloud

500-asset tag estimate: (120 lifts × 22) + (250 ATU res × 10) + (130 ATU com × 15) ≈ **2,640 + 2,500 + 1,950 = ~7,090 tags** — set `MOOREVIEW_MAX_TAGS=8192` for wastewater tenant.

Data ingest @ 500 field assets

Lift stations run **higher-rate PdM** (cloud storage index **1.4×** per Infrastructure Projections). ATU sites are moderate.

Layer	500 assets × 50 MB/mo (weighted avg)	32 WWTP × ~100 MB/mo	Combined
Monthly ingest	~ 25 GB	~ 3.2 GB	~ 28 GB
Hot Mongo (7-day)	~5.8 GB	~0.7 GB	~ 6.5 GB
Archive Year 1	~300 GB	~38 GB	~ 338 GB

Archive volume: start **200 GB** (\$20/mo), expand to **~350 GB** (~\$35/mo) by month 12.

Scale triggers (wastewater fleet)

Signal	Threshold	Action
Fleet devices	> 1,000	Second MQTT broker or tenant shard
Tags	> 8,192 per tenant	Raise <code>MOOREVIEW_MAX_TAGS</code> ; scale Mongo compute
Archive	> 500 GB	Expand block volume; evaluate object storage
Lift PdM ingest	> 40 GB/mo fleet	Step SaaS to 8 GB droplet; Mongo 50 GB tier
WWTP appliances	> 50 head-ends	Dedicated site-agent relay or regional MQTT

Part 4 — Financial model

3-year revenue (wastewater vertical)

Directional model — **wastewater-focused** channel (ACE + utility + OEM). Aligns with portfolio wastewater benchmark.

Metric	Year 1	Year 2	Year 3
New assets installed	~230	~150	~120
Cumulative assets	~230	~380	~ 500
One-time (install + commissioning)	~\$460K	~\$375K	~\$300K
Renewal + cell (in-year + ARR build)	~\$90K	~\$365K	~\$600K
Total revenue	~ \$550K	~ \$1.74M	~ \$3.9M
End-of-year renewal ARR	~\$55K	~\$175K	~\$400K

Year 2–3 assume 10–15% fleet renewal discount on operators above 50 sites; WWTP head-ends at commercial tier ACV.

Year-1 site mix (wastewater launch)

Segment	Tier	Y1 target	Y1 revenue (list)	Y1 recurring added
Lift station — duplex	Commercial	40	\$113,440	\$13,440
ATU — residential	Residential	100	\$175,200	\$25,200
ATU — commercial	Commercial	60	\$170,160	\$20,160
WWTP / package plant head-end	Commercial	8	\$22,688	\$2,688
Wells & pumps (adjacent)	Commercial	22	\$62,392	\$7,392

Total		~230	~\$543,880	~\$68,880 ARR
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Rounded to ~**\$550K** year-1 total with in-year renewal recognition and OEM NRE.

Steady-state fleet @ 500 assets (Year 3 end)

Reference mix for Phase 1 capacity and infra cost model:

Segment	Count	Renewal/mo	Monthly renewal
Lift station — duplex	120	\$25	\$3,000
ATU — residential	250	\$18	\$4,500
ATU — commercial	100	\$25	\$2,500
WWTP head-end (no per-asset renewal)*	32	—	—
Total billable assets	470		\$10,000

WWTP head-ends billed as commercial sites (\$25/mo) in operator contracts; 32 appliances share tenant fleet view. Full 500-device MQTT count = 470 field + 32 head-end + overhead.

With 32 WWTP @ \$25/mo: total renewal ~~\$10,800/mo~~ (\$129,600/yr).

Part 5 — Infrastructure & cellular cost (2026)

Actual carrier costs (planning basis)

Fleet class	SIM	Data plan	Cost/asset/mo	List to customer
Field assets (lift, ATU, IoT-Link)	\$2.00	\$0.50 / 50 MB	\$2.50	\$3.00
WWTP head-end appliances (cameras, site agent)	\$2.00	\$5.00 / 1 GB	\$7.00	Pass-through or bundled

50 MB/mo per lift/ATU is sufficient for 5 s Parc telemetry + alarm events. 1 GB/mo supports site-agent heartbeat, inventory sync, and occasional camera proxy — not continuous HD streaming.

Cloud infrastructure (Phase 1 @ 500 assets)

Component	Monthly	Annual
SaaS 4 GB droplet	\$24	\$288
MQTT 2 GB droplet	\$12	\$144
Archive 2 GB droplet	\$12	\$144

Archive volume (~338 GB Y1 avg)	~\$34	~\$408
Managed Mongo (15 GB base)	\$15.23	\$183
Cloud subtotal	~\$97	~\$1,167

Cellular @ 500 + 32

Line	Calculation	Monthly	Annual
Field fleet (500 × \$2.50)	SIM + 50 MB	\$1,250	\$15,000
WWTP head-ends (32 × \$7.00)	SIM + 1 GB	\$224	\$2,688
Cellular subtotal		\$1,474	\$17,688

Combined P&L — infrastructure layer (Year 3 steady state)

	Monthly	Annual	% of ~\$10,800 renewal
Renewal revenue (500 assets, list)	\$10,800	\$129,600	100%
DO Phase 1 cloud	−\$97	−\$1,167	0.9%
Cellular — field (cost)	−\$1,250	−\$15,000	11.6%
Cellular — WWTP (cost)	−\$224	−\$2,688	2.1%
Total infrastructure (cost)	−\$1,571	−\$18,855	14.5%
Contribution after infra cost	\$9,229	\$110,745	85.5%

Cell pass-through margin

	Monthly
Cell billed (500 × \$3 list)	\$1,500
Cell cost (500 × \$2.50)	−\$1,250
Cell margin	\$250
WWTP cell (32 × \$7 cost; often bundled in site fee)	−\$224

Per-asset economics (commercial lift station)

	Per asset/mo	Per asset/yr
Renewal revenue	\$25.00	\$300.00
Cell billed (list)	\$3.00	\$36.00

Cell cost	\$2.50	\$30.00
Cloud infra (allocated, 500 fleet)	~\$0.19	~\$2.33
WWTP uplink (allocated)	~\$0.04	~\$0.54
Contribution after infra	~\$22.27	~\$267.13

Gross margin on renewal + cell (list): $(\$25 + \$3 - \$2.50 - \$0.23) / \$28 \approx 89\%$ before sales, install labor, and hardware COGS.

Infrastructure cost by year (ramp)

Year	Cumulative assets	Cloud/mo	Cell/mo (field)	WWTP cell/mo	Total infra/mo
Y1	~230	~\$93	~\$575	~\$112 (16 units)	~\$780
Y2	~380	~\$95	~\$950	~\$168 (24 units)	~\$1,213
Y3	~500	~\$97	~\$1,250	~\$224 (32 units)	~\$1,571

Phase 1 stack is **largely fixed** from launch — margin improves with fleet density until archive/Mongo tier step-up (~600+ high-rate lift assets).

Break-even (infrastructure only)

Cost covered	Assets @ \$25/mo renewal
Cloud stack alone (~\$97/mo)	~4 assets
Cloud + field cell (~\$1,347/mo)	~54 assets
Full stack incl. 32 WWTP (~\$1,571/mo)	~63 assets

Part 6 — Unit economics & targets

Metric	Target
Renewal gross margin (after cloud + cell cost)	85 – 90%
Hardware + commissioning margin	35 – 45%
Blended year-1 payback on install COGS	Under 14 months
Net revenue retention (fleet operators)	> 110%
Logo churn (annual)	< 5%
LTV (commercial lift, 7 yr)	$\$2,500 + (\$336 \times 7) \approx \$4,852$

LTV:CAC target > 3:1 on contractor-led installs. ACE channel reduces CAC vs cold national sales.

Part 7 — Go-to-market

Phase roadmap

Phase	Assets	Focus	Cloud
Launch (Y1)	~230	ACE reference fleet; lift + ATU routes	Deploy Phase 1; 10 GB Mongo
Growth (Y2)	~380	Utility pilots; OEM panel embed	Archive expand 200→300 GB
Scale (Y3)	~500	Fleet discounts; municipal RFP	Phase 1 at capacity; plan ingest shard

Sales channels (wastewater)

Channel	Y1 mix	Advantage
Purple Standard / ACE	40 – 50%	Install crews, brand, first 500 sites
Contractor monitoring contracts	30 – 40%	Attach on PM routes
OEM / panel embed	10 – 15%	ATS panel + provider Parc
Utility / municipal	5 – 10%	Longer cycle; 13 mo compliance archive

Compliance & retention drivers

- 13-month historian archive on lift/ATU (regulatory route documentation)
- PdM → proactive CMMS — service due before inspection failure
- Portable .est — contractor owns project; no vendor lock-in on recommission

Part 8 — Risk & scale path

Risk	Mitigation
Cellular outage at lift station	Local alarm logic on Opta; spool on IoT-Link; SSO dialer as last line
MQTT broker load	Dedicated Phase 1 MQTT droplet; lock pilot IPs; scale at 2K connections
Archive growth (lift PdM)	7-day hot → zstd; expand block volume \$0.10/GiB
Tenant tag overflow	Size MOOREVIEW_MAX_TAGS=8192 for wastewater; monitor dashboard
National scale (>5K assets)	VM split: ingestion · alarms · GUI · AI (ARCHITECTURE.md)

Part 9 — Comparison to Product Market Entry (portfolio)

This vertical plan vs **MooreVIEW-Product-Market-Entry.md v1.5** (full portfolio launch).

Revenue trajectory

Metric	Wastewater (this plan)	Portfolio market entry
Y1 total revenue	~\$550K	~\$550K – \$600K
Y2 total revenue	~\$1.74M	~\$1.5M
Y3 total revenue	~\$3.9M	~\$3.5M
Y1 assets/sites	~230 WW	237 mixed (95 WW = 40%)
Y3 cumulative	~500 WW	~1,487 all verticals
Y3 renewal ARR	~\$400K	~\$780K

Wastewater-only **concentrates revenue** in fewer assets with **higher Y2–Y3 totals** than the blended portfolio — because ACE/Purple Standard channel and utility pilots weight lift + ATU before pools/HVAC scale.

Infrastructure & Phase 1 cloud

Dimension	Wastewater @ 500	Portfolio @ 1,487 sites
Deployment model	Cloud-centric (IoT-Link)	Mostly cloud-centric + 2 ALF Y1
Phase 1 device load	500 MQTT — at stack capacity	~1,487 sparse — same stack OK
Cloud tags (est.)	~7,090	~9,000 + ALF rollups
Hot Mongo (7 d)	~6.5 GB	~15 – 60 GB (Y1 band)
Dominant opex	Cellular \$1,250/mo	Cellular + archive growth
Infra % of renewal	~15% @ WW list pricing	~15 – 31% by mix
Cloud stack cost	~\$97/mo	~\$97 – 145/mo (Y1 – Y3)

Pricing alignment

Both documents use **Pricing Guide v2.4** list: commercial **\$2,836 Y1 · \$336/yr** recurring with cell. This plan adds **actual carrier cost** (\$2.50/mo field) for margin planning; Product Market Entry uses **\$3/mo list** cell line item.

What portfolio includes that WW omits

Vertical	Y1 sites (portfolio)	In WW plan?
Pools (res + com)	45	No

HVAC (res + com)	70	No
Convenience store	10	No
ALF / campus	2	No (see ALF scaling in Infrastructure Projections Part 10)
Wells (adjacent)	15	Partial (22 in WW Y1 mix)

Strategic positioning

	Wastewater vertical	Portfolio
Beachhead	ACE Septic · lift + ATU routes	Lift + c-store + pools + HVAC
Phase 1 stress test	500 devices on one MQTT broker	Device count higher; telemetry sparser
5-yr ARR ceiling	\$15 – 25M (WW SAM)	\$50 – 120M (all verticals)
When to use this doc	Utility RFP · ACE fleet · WWTP OEM	Investor · multi-vertical GTM

Appendix — assumptions

Item	Value
Fleet reference	500 field assets + 32 WWTP head-ends
Field cellular cost	\$2/SIM + \$0.50 per 50 MB/mo
WWTP cellular cost	\$2/SIM + \$5 per 1 GB/mo
Field cellular list	\$3/mo (Pricing Guide v2.4)
Cloud topology	Phase 1 NYC1 (SaaS + MQTT + Archive + Mongo)
Revenue benchmarks	Wastewater vertical plan Y1–Y3
Tag estimate	~7,090 @ 500 assets; cap 8,192
Ingest	50 MB/mo/field asset; ~100 MB/mo/WWTP head-end
DO pricing	Aug 2026 list

Disclaimer: Directional business plan for capacity, pricing, and opex review. Not a financial forecast, hardware quote, or SLA commitment.

Related documents

Document	Purpose
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Phase1-SaaS-Architecture-Costs.md	Phase 1 topology + generic 500-device cost model
DO-Phase1-Storage-Cost-Review.md	Storage-only deep dive
MooreVIEW-Pricing-Guide.md	List MSRP v2.4
MooreVIEW-Product-Market-Entry.md	Portfolio launch model v1.5 (comparison Part 9)
MooreVIEW-Infrastructure-Projections.md	5-year scaling · Part 9 Phase 1 · Part 10 ALF
CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md	Production deploy runbook

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